Week 1 Notes

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1 Learning Unit 3-1: Conversions

1.1 Percents

Like fractions and decimals, **percents** describe parts of a whole. The word *percent* means "per 100", and the percent symbol (%) indicates hundreths (division by 100). Percents are the result of expressing numbers as part of 100.

1.2 Converting Decimals to Percents

Step 1 Move the decimal point two places to the right. This represents multiplying by 100. If necessary, add zeros. This rule is also used for whole numbers and mixed decimals.

Step 2 Add the percent symbol at the end of the number.

1.2.1 Examples

- $0.49 = 49$
- $0.80 = 80$
- $8.00 = 800$
- $0.007 = 0.7$

1.3 Rounding Percents

The rules for rounding percents isn't necessarily universal, but these will be the rules for this class:

Step 1 Only round once you a percentage. Make sure to convert from fractions or decimals first.

Step 2 When rounding, you want the digit to the right of the digit in question to be 5 or greater. Then, you can round that identified digit up.

Step 3 Delete digits to the right of the identified digit.

1.3.1 Examples

- $\frac{18}{55} = 32.727272\ldots$ **rounded**: 32.73
- $\frac{2}{3} = .66666666\ldots$ **rounded**: $.67$

1.4 Converting Percents to Decimals

To convert percents to decimals, just reverse what we learned for converting decimals to percents.

Step 1 Drop the percent symbol

Step 2 Move the decimal point two places left. This is the same as dividing by 100. If necessary, add zeros.

1.5 Converting Fractions to Percents

When fractions have denominators of 100, it's fairly simple to convert it a percent. If it doesn't here are the steps.

Step 1 Divide the numerator by the denominator to convert the fraction to a decimal.

Step 2 Move the decimal point two places to the right, don't forget to add the percent sign.

1.6 Converting Percents to Fractions

You can write any percent as a fraction whose denominator is 100.

Step 1 Drop the percent symbol.

Step 2 Multiply the number by $\frac{1}{100}$.

Step 3 Reduce to lowest terms.

1.6.1 Example

Convert 76% to a fraction.

Step 1 $76\% = \frac{76}{100}$

Step 2 **Reduce**: $\frac{76}{100} = \frac{19}{25}$

1.7 Converting a Mixed or Decimal Percent to a Fraction

Step 1 Drop the percent symbol.

Step 2 Change the mixed percent to an improper fraction.

Step 3 Multiply the number by $\frac{1}{100}$

1.7.1 Example

Convert $12\frac{1}{2}$ % to a fraction:

Step 1 Convert $12\frac{1}{2}$ % to an improper fraction.

- $(2 * 12) + 1 = 25$

- $\frac{25}{2}$

Step 2 Multiply $\$25\frac{}{2*\frac{1}{100}}$

- **Equals**: $\frac{25}{200}$

Step 3 **Reduce**: $\frac{25}{200} = \frac{1}{8}$

2 Learning Unit 3-2: Application of Percents-Portion Formula

Base *(B)* The **base** is the beginning whole quantity or value (100%).

Rate *(R)* The **rate** is a percent, decimal, or fraction that indicates the part of the base that you must calculate. Look for the percent sign to identify the rate.

Portion *(P)* The **portion** is the amount or part that results from $base * rate$.

2.1 Solving Percents with the Portion Formula

2.1.1 TODO practice these word problems!!!!